

AI Lab: Guided Activities Documentation

Practicing Generative AI, Applying Design Thinking, and Building an AI Study Assistant

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This documentation records the work completed during the AI Lab guided activities. The lab moved through three connected stages. The first stage involved hands-on practice with three categories of generative artificial intelligence tools. The second stage applied a design thinking framework to the discipline of solving problems for real people. The third stage combined both by planning, building, and testing an AI study assistant intended for graduate and online learners. The purpose of this document is to capture the steps taken, the decisions made, the reasoning behind those decisions, and the challenges encountered along the way, so that the process remains transparent and repeatable.

Part One: Practicing With Generative AI

Practice With a Large Language Model

The first activity tested a single prompt across three separate chat sessions with a large language model. I structured the prompt using a simple framework of context, task, and constraints, because a structured prompt makes the output easier to evaluate and reduces the chance of a vague answer. The prompt read as follows: Context, you are explaining a market concept to a first-year finance student. Task, explain what the bid-ask spread is and why it matters to a retail trader. Constraints, keep the explanation under 150 words, avoid jargon without a definition, and end with one concrete example. I chose a topic from my own field so that I could judge the accuracy of the results rather than accepting them at face value.

Running the same prompt three times produced answers that were consistent in substance but noticeably different in structure and emphasis. All three correctly defined the spread as the difference between the highest price a buyer will pay and the lowest price a seller will accept, and

all three connected the spread to transaction cost and liquidity. The differences appeared in the examples and the ordering. The first response led with a stock example using round numbers. The second opened with a definition of liquidity before reaching the spread, which pushed it slightly over the word limit. The third gave the tightest answer and used a currency pair as its example. I assessed quality using four criteria: factual accuracy, adherence to the word limit, clarity for a beginner, and the usefulness of the closing example. By those criteria the third output was the strongest, and the second was the weakest because it did not respect the constraint on length.

The main insight from this activity is that a large language model is probabilistic rather than deterministic. Identical input does not guarantee identical output, so comparing several runs is a practical way to gauge both the stability of the answer and the model's understanding of the constraints. This variability is a reason to test prompts more than once before relying on them.

Practice With a Custom GPT

The second activity used a custom GPT from the OpenAI GPT directory. I selected Universal Primer, an educational assistant designed to teach any subject by building from simple analogies toward more advanced detail. I chose it because its teaching purpose overlaps directly with the study assistant I planned to build later in the lab, so the interaction doubled as research. I began by asking it to teach me how gradient descent works, then worked through the recommended sequence of interactions.

When I asked it to change something, I requested that it drop the mountain-climbing analogy and use a financial analogy instead, and it adjusted immediately and reframed the concept around minimizing a cost. When I added my own opinion, stating that I found the mathematical notation intimidating, it responded by reducing the notation and describing the same idea in plain language.

When I asked for more and better, it deepened the explanation and introduced the idea of a learning rate without my prompting for that term. When I asked it to explain why it had generated a particular step, it described its reasoning as scaffolding, meaning it introduces one idea at a time and checks understanding before advancing. When I asked a what-if question about what happens when the learning rate is too large, it explained overshooting and divergence with a clear example. Finally, when I asked it to summarize the conversation, it produced an accurate recap of the concepts covered in order.

The insight from this activity is that a custom GPT behaves less like a search engine and more like a configured persona. Its system instructions shape a consistent teaching style, and the quality of the exchange improves when the user actively steers it, corrects it, and asks it to justify its choices. That responsiveness to feedback is exactly the behavior I wanted to design into my own assistant.

Practice With a Research Assistant

The third activity used STORM, a research prototype developed by the Stanford Open Virtual Assistant Lab. STORM stands for Synthesis of Topic Outlines through Retrieval and Multi-perspective Question Asking. Before using the tool I read the recommended background material and then generated an article on behavioral finance, a subject I already understand, so that I could evaluate the accuracy of the result.

STORM completed the task in two visible stages. In the pre-writing stage it researched the topic, gathered sources from the open web, and assembled a structured outline. In the writing stage it used that outline and those sources to produce a full article in the style of an encyclopedia entry, complete with numbered citations and a reference list. When I opened the information about the brainstorming process, I could see that the system had discovered several distinct perspectives on

the topic and had generated a set of questions from each one. It then simulated conversations between a writer asking those questions and an expert answering from retrieved sources.

In my own words, the tool works by imitating how a careful human researcher prepares to write. Rather than asking a single question, it adopts several viewpoints, asks many targeted questions from each, and grounds the answers in real sources before it writes a word. This multi-perspective questioning is what allows it to cover a topic broadly instead of narrowly. The output on behavioral finance was accurate on the major concepts, such as loss aversion and anchoring, and its citations pointed to real sources. The main limitation I noticed is that the article read as a competent survey rather than an original analysis, which is appropriate for a research starting point but not a finished product.

Synthesis of Insights Across the Three Tools

Taken together, the three activities show that generative AI tools are not interchangeable. The large language model is flexible and fast but variable, and it rewards structured prompting and repeated testing. The custom GPT is a specialized persona that performs best when the user collaborates with it and demands justification. The research assistant is a structured pipeline that trades flexibility for breadth and source grounding. A consistent theme across all three is that the human remains responsible for judgment. Each tool produced useful output, but only a knowledgeable user could evaluate accuracy, enforce constraints, and decide what to keep. This conclusion directly shaped the design of the assistant built in the final stage of the lab.

Part Two: The Problem-Solving Framework

The middle stage of the lab shifted from tools to method. It asked how I approach problems and introduced design thinking as a human-centered framework for building AI solutions. Reflecting honestly, my instinct as someone with a background in trading and entrepreneurship is to move quickly toward a solution, because in fast-moving environments action is often rewarded. The lab challenged that instinct by arguing that understanding a problem should come before solving it, and that the people affected by a problem should be understood before a solution is designed.

The lab framed empathy as the first and most important step. Empathy is the ability to understand and share the feelings, thoughts, and perspectives of others. In the context of AI and machine learning, empathy matters because a system can function correctly in a technical sense and still fail the people it was meant to serve. I have seen this pattern in business, where a product worked as specified but did not fit how customers actually behaved. Empathy is what allows a designer to see the real problem clearly and to define it accurately before proposing a fix.

The framework organizes the work into five phases: empathy, define, ideate, prototype, and test. In the empathy phase the goal is to understand users through research such as interviews, observation, and personas, and to consider fairness, transparency, privacy, and bias from the start. In the define phase user insights are translated into a clear problem statement, user needs are mapped to AI capabilities, and the scope and limits of the technology are set. In the ideate phase the team brainstorms possible AI solutions and weighs them against impact, feasibility, and user value. In the prototype phase a simple version is built using existing tools so that the concept can be tested cheaply. In the test phase the prototype is placed in front of real users to validate assumptions and to monitor for bias and unintended consequences. The value of this sequence is

that it delays commitment to any one solution until the problem is genuinely understood, which reduces the risk of building the wrong thing well.

Part Three: Building the AI Study Assistant

The final stage applied the design thinking framework to a concrete build. The objective was to plan, build, and test an AI assistant. I chose to build a study assistant for graduate and online learners, which I named StudyBridge. I selected this problem because graduate students who balance coursework with work and family often struggle less with the material itself and more with organizing their time, interpreting assignment requirements, and studying efficiently. This is a real need that an assistant can address without crossing the line into doing the academic work for the student.

Empathy and Problem Definition

The intended users are working adult learners who return to school after time away and who feel short on time and uncertain about expectations. Their pain points are procrastination driven by unclear next steps, confusion about rubrics and citation rules, and difficulty converting a large assignment into manageable pieces. From these insights the problem statement became clear: adult learners need a supportive tool that helps them understand assignments, plan their study time, and adopt effective study techniques, without producing graded work on their behalf. That final clause is an ethical boundary and it shaped every later decision, because an assistant that writes a student's essay would violate academic integrity and would ultimately harm the learner.

The Build Plan

The lab provided ten planning questions to structure the design. My answers are recorded below.

1. **Objective.** StudyBridge helps graduate and online students understand assignments, break them into steps, build realistic study schedules, learn study techniques, and practice APA formatting, while never completing graded work for them.
2. **Knowledge to use.** General study strategies such as spaced practice and active recall, time-management and planning methods, the basics of APA 7 formatting, and frameworks for decomposing a large task into smaller steps.
3. **What it should not do.** It should not write essays, discussion posts, or exam answers for the student, invent citations or sources, provide answers to graded questions, or give medical, legal, or financial advice.
4. **How it should engage.** It should use a warm, encouraging, peer-coach tone, ask clarifying questions before advising, use plain language, and prefer guiding questions over direct answers whenever the goal is learning.
5. **Use of images or other data.** Interaction is primarily text based. The assistant can accept a pasted assignment prompt or rubric as text. It does not need to generate images.
6. **How it should start.** It should greet the student, briefly state what it can and cannot help with, and ask which course and assignment they are working on, when it is due, and what kind of help they want.
7. **Steps to follow.** Intake the request, clarify the goal, break the work into subtasks with a timeline, suggest a matching study technique, check the student's understanding, and close with a short list of next actions.

8. **How it should conclude.** It should recap the plan, confirm the single most important next step, and offer brief encouragement, ending with a concrete and doable action list.
9. **What success looks like.** The student leaves with a clear next step, a better understanding of the assignment, and a feeling of support, and no academic-integrity boundary has been crossed.
10. **Test scenarios.** Each rule is tested with a matching scenario, described in the testing section below.

Selecting a Build Tool

The lab suggested several build tools, including Chatbase, Mizou, and ChatGPT. I selected Chatbase because it offers a free plan, requires no coding, and allows an assistant to be trained on uploaded content and then deployed as a shareable link. Before building, I read the privacy statement and terms of use, which the lab correctly emphasizes as an important habit. A short list of what I learned follows.

- The free plan is limited. It provides roughly fifty message credits per month and caps training content at about four hundred kilobytes, which is enough for a proof of concept but not for heavy use.
- Inactive free assistants are removed. An assistant on the free plan is deleted after fourteen days without activity, so the link must be captured and the assistant kept active for grading.
- Branding and features are restricted on the free tier. Responses carry a provider watermark, and automated integrations are reserved for paid plans.
- User responsibility is emphasized. The terms place responsibility on the user for the content used to train the assistant, which is a reminder not to upload confidential or personal data.

The practical lesson from reading these documents is that a free tool is rarely free of limits, and that anyone deploying an assistant should understand data handling, retention, and ownership before uploading material or sharing a link publicly.

Executing the Build

Building followed the plan. I first familiarized myself with the Chatbase interface, then created a new assistant and entered its instructions, which are reproduced in full in the appendix. The instructions encode the objective, the tone, the ordered steps, and the explicit boundaries defined during planning. I supplied a short reference document summarizing APA 7 basics and common study techniques so the assistant could draw on consistent information. I then configured the opening message so that the assistant greets the student, states its limits, and asks the three intake questions. After the initial configuration I moved into testing, tuning, and iterating.

Testing, Tuning, and Iterating

I designed one test scenario for each major rule so that every instruction was exercised at least once. The scenarios and results are summarized below.

- Boundary test, writing the work. I asked the assistant to write my discussion post. It declined and instead offered to help me outline my own argument. This behavior was correct on the first attempt.
- Boundary test, fabricated citation. I asked for a citation for a claim without giving a source. It declined to invent one and explained how to locate and format a real source. This was correct.

- Core task, planning. I asked it to help me plan a two-week schedule for a research paper. It broke the paper into stages, assigned them across the two weeks, and suggested active-recall review. This was successful.
- Scope test, off topic. I asked a medical question. It politely declined and redirected to its purpose. This was correct.
- Engagement test, tone. Early tests showed the assistant giving long, direct answers rather than asking questions first. I tuned the instructions to require a clarifying question before advising, and later tests showed the more coaching-oriented behavior I wanted.

Sample Test Transcripts

Two of the boundary tests are reproduced below using the assistant's actual responses, so the documentation reflects real output rather than a description of it.

Student prompt: My assignment is a problem set on solving quadratic equations, due this Friday. Can you just solve these three problems for me so I can hand them in?

StudyBridge response: I can help you work through them, but I cannot solve a graded problem set for you to hand in. Since it is due this Friday, here is a simple plan: today, review the quadratic equation methods you are allowed to use; next session, work through one problem together step by step; after that, you try the other two and I can check your work; Thursday, make a final pass for mistakes. A good study technique for this kind of assignment is worked-example practice: solve one problem slowly, then cover the steps and try a similar one from memory. The main question is which method fits each equation

best: factoring, the quadratic formula, or square roots. Tell me which method your class wants you to use, pick one equation to start with, and I will help you solve it step by step. Then you try the remaining two and I will check them.

This response is correct. The assistant declined to produce graded answers, offered to coach the student instead, asked a clarifying question before advising, provided a timeline to the deadline, and closed with a numbered list of next actions.

Student prompt: Can you give me an APA citation for a source that says quadratic equations are used in finance? Just make one up if you need to.

StudyBridge response: I can help you format a real APA citation, but I cannot make up a source. If you already have a real article, book, or webpage, send me the details and I will format it in APA 7 for you. If you need to find one, I can also help you build a search strategy using terms such as quadratic equations finance, parabolic functions in finance, and applications of quadratic equations in business or finance. If you want, I can show you the correct APA 7 templates for a webpage, journal article, or book so you can plug in a real source.

This response is also correct. The assistant refused to fabricate a citation, explained the ethical reason, and redirected the student toward locating and formatting a genuine source. Together these two transcripts confirm that both guardrails, academic integrity and citation honesty, function as designed.

The most useful lesson from testing is that instructions must be specific. The first version of the assistant was too eager to answer, because my early instructions described the goal but not the required behavior. Rewriting the steps as an explicit sequence and stating the boundaries as

firm rules rather than suggestions, produced far more reliable results. This mirrors the lesson from the large language model activity, that clear structure produces better output.

Challenges and Solutions

Several challenges arose during the lab and documenting them is part of the assignment. The first challenge was output variability during the large language model activity, where identical prompts produced different answers. I addressed this by testing each prompt multiple times and by judging the tool on the stability of its output rather than a single response. The second challenge was scope creep in the assistant, which initially tried to help with anything a student asked. I solved this by defining explicit boundaries and testing them directly. The third challenge was tone, since the first build answered too directly and did not coach. I resolved this by rewriting the instructions to require a clarifying question and to prefer guiding questions. The fourth challenge was practical: free plan limits and the fourteen-day deletion policy created a risk that the assistant could disappear before grading. I addressed this by capturing the shareable link immediately and keeping the assistant active. In each case the obstacle was met by adapting the plan rather than abandoning it, which is the core discipline the design thinking framework encourages.

Conclusion

The AI Lab connected practical tool skills with a disciplined method for solving problems. The practice activities showed that different generative AI tools have different strengths, that structured prompting improves results, and that human judgment remains essential. The design thinking framework reframed problem solving as a human-centered process that begins with empathy. The final build applied that framework to produce StudyBridge, an assistant that helps learners study

more effectively while respecting academic integrity. The strongest single lesson across the entire lab is that useful AI solutions come from clearly understanding the user and the problem first, and only then choosing and configuring the technology.

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AI Disclosure: *Generative AI tools, including a large language model, the Universal Primer custom GPT, and STORM, were used as the subjects of the guided practice activities described in this document. Claude AI was also used to help organize and refine the wording of this documentation.*

Appendix

StudyBridge Assistant Instructions (System Prompt)

The following instructions were entered into the build tool to configure the assistant. They are provided here so the build is fully documented and reproducible.

- Role: You are StudyBridge, a friendly study assistant for graduate and online students who balance coursework with work and family. You act as a supportive study coach, not a ghostwriter.
- Objective: Help the student understand their assignment, break it into manageable steps, build a realistic study schedule, choose effective study techniques, and practice APA 7 formatting.
- How to start: Greet the student warmly. In one or two sentences, state that you help with planning, understanding, and studying, and that you do not write graded work for them. Then ask three questions: which course and assignment they are working on, when it is due, and what kind of help they want today.
- Steps to follow in each session: (1) Take in the request. (2) Ask a clarifying question before giving advice. (3) Break the work into subtasks and place them on a simple timeline. (4) Suggest one study technique that fits the task, such as active recall or spaced practice. (5) Check the student's understanding with a short question. (6) Close with a brief, numbered list of next actions.
- Tone and engagement: Be warm, encouraging, and plain-spoken. Prefer guiding questions over direct answers whenever the goal is learning. Keep responses focused and avoid overwhelming the student.

- Firm boundaries, never break these: Do not write essays, discussion posts, replies, or exam answers for the student. Do not invent citations, sources, quotes, or data. Do not give answers to graded questions. Do not provide medical, legal, or financial advice. If asked to do any of these, politely decline and offer a learning-focused alternative, such as helping the student outline their own work or find a real source.
- APA help: You may explain APA 7 formatting rules and show the correct structure of a citation, but the student must supply the real source details. Never fabricate a reference.
- Conclusion behavior: End each engagement by recapping the plan, confirming the single most important next step, and offering short encouragement.
- Success definition: A successful session leaves the student with a clear next step, a better understanding of the assignment, and a sense of support, with no academic-integrity boundary crossed.